How long can a two-cycle quantum conveyor operate?

ResearchPhO — Solutions and marking scheme **30 points**

How to use this booklet

Each answer is followed by its marking scheme; the numbers in a marking line add up to the points of the question. Award method marks for a consistent derivation, partial credit for correct approximate reasoning, and no credit for a claimed pump lifetime when only a toy-model oscillation, a source norm or a short-time coefficient has been obtained. In a marking line, (M) marks a modelling step (setting up the model, choosing the physics, interpreting a result, judging its scope) and (C) a computation step (algebra, solving, evaluating). Appendices A–C at the end contain the research context that does not belong in the question paper: the research map, the source and scope notes of every question, and teacher notes. Appendix D, the toolbox, works the smallest example of each tool that the question paper states without proof.

Mark allocation

Problem	Questions	Points
1	A.1 (1.5), B.1 (2.7), B.2 (1.4), C.1 (2.3), D.1 (2.1)	10.0
2	A.1 (1.3), B.1 (1.8), B.2 (0.8), C.1 (1.8), C.2 (1.3), D.1 (0.9), D.2 (2.1)	10.0
3	A.1 (1.1), A.2 (2.2), A.3 (2.0), B.1 (1.6), C.1 (1.7), C.2 (1.4)	10.0
		30.0

Problem 1: A two-cycle quantum conveyor: bands, projectors and flavors

(10.0 points)

Part A. Fractional transport

(1.5 points)

A.1 Two-cycle particle counting

1.5 pt

From the supplied displacement rule,

$$Q(0,2T) = \frac{1}{Na}\sum_{j=1}^{N}(-a) = -1, \qquad \boxed{\overline{Q} = -\frac{1}{2}.}$$

Only the sum of the two single-period charges is fixed. For example,

$$Q_1 = q, \qquad Q_2 = -1 - q$$

is consistent for any allowed q. The result is an expectation-value transport statement. It does not require a fractional particle in a single measurement.

Marking: Two-period charge: 0.75 (C). Average: 0.45 (C). Unequal single-period values and interpretation: 0.3 (M).

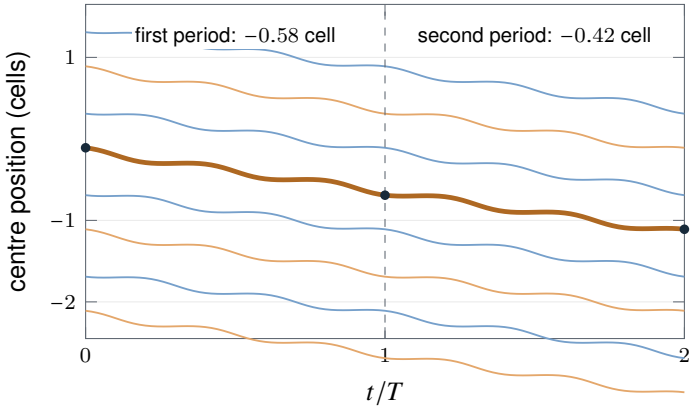


Figure 1.1. A.1 — only the two-period sum is fixed. The followed orbital of Figure 1.3(b) of the question paper (single-particle geometry at $K/J = -3/2$). Its first period moves it by about -0.58 cell and its second by about -0.42 cell. Only the sum, -1, is fixed by the two-cycle return, so $Q(0,T)$ and $Q(T,2T)$ need not each equal $-1/2$.

Part B. From the lattice to a projector

(4.1 points)

B.1 Fourier transformation does not solve the interactions

2.7 pt

(a) The seam term becomes

$$\sum_R c^\dagger_{R+1,0} c_{R,4} = \sum_k e^{-ik} c^\dagger_{k,0} c_{k,4}.$$

Thus $H_0 = \sum_k \boldsymbol{c}_k^\dagger h(k,\phi)\boldsymbol{c}_k$, with

$$\boxed{h(k,\phi) = -\begin{pmatrix} 0 & t_0 & 0 & 0 & t_4 e^{-ik} \\ t_0 & 0 & t_1 & 0 & 0 \\ 0 & t_1 & 0 & t_2 & 0 \\ 0 & 0 & t_2 & 0 & t_3 \\ t_4 e^{ik} & 0 & 0 & t_3 & 0 \end{pmatrix}.}$$

For real k, ϕ the hoppings are real and the seam entries are conjugate, so $h = h^\dagger$.

(b) From $Y^\dagger Y = \mathbb{1}_2$,

$$p_{12} = YY^\dagger, \qquad p_{12}^2 = p_{12}, \qquad \operatorname{Tr} p_{12} = 2.$$

It is a rank-two single-particle projector in $\mathbb{C}^5$, or a point of $\mathrm{Gr}(2,5)$. The source's composite Wannier mechanism motivates choosing these bands. The choice is not made because a rank-two calculation is convenient.

(c) The three spaces are different:

Restriction	Dimension at $N = 4$
All $5N$ orbitals; only total particle number fixed	$\binom{20}{4} = 4845$
Only the two composite bands; total particle number fixed	$\binom{8}{4} = 70$
Two flavor orbitals per cell; exactly one occupied in every cell	$2^4 = 16$

Translation-sector restrictions can reduce a dimension further; they are not included in this table. The single-particle Fourier transform leaves the physical interaction as a many-body operator. It does not make the full dynamics a 5×5 problem.

Marking: (a) Matrix from the lattice, with both seam entries and Hermiticity: 1.05 (M). (b) Rank two in $\mathbb{C}^5$: 0.35 (C); physical motivation: 0.4 (M). (c) The three restrictions: 0.35 (M); the three counts at $N = 4$: 0.25 (C); Fourier transformation leaves the interaction many-body: 0.3 (M).

B.2 The projector generator **1.4 pt**

Differentiate $p^2 = p$:

$$\dot{p}\,p + p\dot{p} = \dot{p}.$$

Multiply by p on both sides. This gives $2p\dot{p}\,p = p\dot{p}\,p$, hence

$$p\dot{p}\,p = 0.$$

Now

$$\begin{aligned}[[\dot{p}, p], p] &= \dot{p}\,p - 2p\dot{p}\,p + p\dot{p} \\ &= \dot{p}.\end{aligned}$$

Therefore $g_t = i[\dot{p}, p]$ obeys

$$\boxed{[g_t, p] = i\dot{p}.}$$

Since $[\dot{p}, p]$ is anti-Hermitian, g_t is Hermitian. It is one generator of the prescribed subspace motion. It does not assert that the physical Hamiltonian produces the same tracking or internal occupation.

Marking: Differentiate and obtain $p\dot{p}\,p = 0$: 0.6 (C). Generator identity: 0.5 (C). What it determines and what it does not determine about the Hamiltonian: 0.3 (M).

Part C. Following a moving projector **(2.3 points)**

C.1 An exactly solvable moving frame **2.3 pt**

The rotation gives

$$W^\dagger H W = -\frac{\Delta}{2}\sigma_z, \qquad -iW^\dagger \dot{W} = -\frac{\nu}{2}\sigma_y.$$

Thus

$$\boxed{\mathcal{K} = -\frac{1}{2}(\Delta\sigma_z + \nu\sigma_y).}$$

With $\omega = \sqrt{\Delta^2 + \nu^2}$,

$$e^{-i\mathcal{K}t} = \mathbb{1}\cos\frac{\omega t}{2} + i\frac{\Delta\sigma_z + \nu\sigma_y}{\omega}\sin\frac{\omega t}{2}.$$

The instantaneous excited state is $W(t)|\downarrow\rangle$, so

$$\boxed{P_{\mathrm{ex}}(t) = \frac{\nu^2}{\Delta^2+\nu^2}\sin^2\!\left(\frac{\sqrt{\Delta^2+\nu^2}\,t}{2}\right).}$$

The path $p(t)$ has no Δ dependence, but the probability does. For $\Delta \gg \nu$ its maximum is small. The frame transformation has not removed the dynamics; it has made that dynamics explicit.

Marking: Moving-frame Hamiltonian: 0.85 (C). Exact probability: 1.05 (C). Same path, different tracking: 0.4 (M).

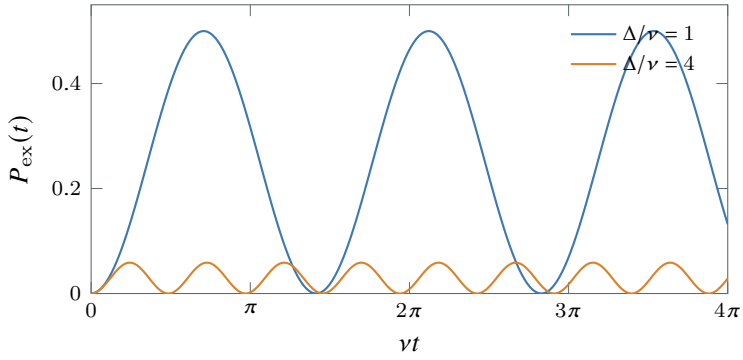


Figure 1.2. C.1 — same projector path, different following. $P_{\mathrm{ex}} = \frac{\nu^2}{\Delta^2+\nu^2}\sin^2\!\left(\frac{1}{2}\sqrt{\Delta^2+\nu^2}\,t\right)$. The path $p(t)$ is identical in both curves; only Δ differs. A larger gap gives a smaller and faster oscillation.

Part D. A plane is not yet a flavor **(2.1 points)**

D.1 A gauge change is not a physical change of flavor **2.1 pt**

(a) Since $\Lambda\Lambda^\dagger = \mathbb{1}_2$,

$$(Y\Lambda)(Y\Lambda)^\dagger = YY^\dagger = p_{12}.$$

For a fixed selected physical direction,

$$(Y\Lambda)(\Lambda^\dagger v_A)(\Lambda^\dagger v_A)^\dagger(Y\Lambda)^\dagger = Yv_Av_A^\dagger Y^\dagger = p_A.$$

A compensated basis change alters only coordinates. Choosing a different direction in the plane without compensation changes the rank-one physical object. Arbitrary mixing of nondegenerate energy eigenvectors also means they are no longer individual energy eigenvectors; the composite space is still the same.

(b) One needs the Bloch eigenvectors, an internal localization prescription, and the physical position operator, including the orbital positions r_s. In a Wilson-loop implementation, the complete loop matrix around the Brillouin zone supplies internal eigenvectors and center phases with the appropriate basis sewing. One must unwrap and track the centers as ϕ changes. Degeneracies require an explicit continuation rule.

A Wilson loop around k at fixed ϕ organizes Wannier centers. A geometric transport matrix along time organizes a chosen moving frame. The actual evolution uses $H(t)$ and its dynamical phases. These matrices must not be identified solely because all are unitary. The full projector family, spatial embedding, and localization rule can in principle define a flavor; endpoint closure alone cannot.

(c) The two supplied maps obey

$$\det \mathbb{1}_2 = 1, \qquad \det(i\sigma_x) = i^2 \det \sigma_x = 1.$$

But $\mathbb{1}_2$ keeps A fixed, whereas $i\sigma_x|A\rangle = i|B\rangle$. Their determinant phases do not distinguish this action. Neither map includes a spatial displacement, so neither calculation alone proves a nonzero transported charge. A.1 required a separate center winding and actual following.

Marking: (a) Both invariances: 0.45 (C); effect of an uncompensated choice: 0.3 (M). (b) Required data; Wilson loop at fixed ϕ versus evolution in time: 0.75 (M). (c) Equal determinants: 0.2 (C); different action, no displacement: 0.4 (M).

Problem 2: Correlations and the physical current (10.0 points)

Part A. Domains in a spin chain (1.3 points)

A.1 A favorable domain need not form 1.3 pt

Each of the two interfaces changes one bond energy from $-J_z$ to $+J_z$. The interface cost is $4J_z$. Each of the ℓ flipped spins gains field energy $2h$:

$$\boxed{\Delta E(\ell) = 4J_z - 2h\ell.}$$

Strict negativity requires $\ell > 2J_z/h$. Thus

$$\boxed{\ell_{\min} = \left\lfloor \frac{2J_z}{h} \right\rfloor + 1,}$$

provided $\ell_{\min} \leq N - 1$. If no such interior segment fits, this two-interface question has no allowed negative-energy segment.

At $g = 0$, the Hamiltonian is diagonal in the σ^z basis. The all-A state is an eigenstate and acquires only a phase. It does not form a domain even when another configuration has lower energy.

Energy preference is not a transition amplitude or a rate.

Marking: Energy change of the domain: 0.45 (M). Integer threshold and fit condition: 0.3 (C). Stationary eigenstate and distinction from a rate: 0.55 (M).

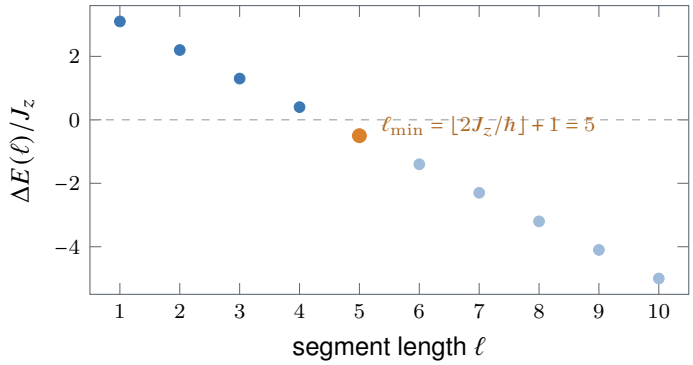


Figure 2.1. A.1 — domain energy (illustrative $J_z = 1$, $h = 0.45$). $\Delta E(\ell) = 4J_z - 2h\ell$ decreases linearly; here $2J_z/h \approx 4.44$. A negative ΔE only says the domain is lower in energy. With $g = 0$ nothing in H_{d} can flip a spin, so the all-A state stays where it is.

Part B. Dressing a reference state (2.6 points)

B.1 A two-channel correlated state 1.8 pt

The characteristic equation is $E^2 - \Delta E - v^2 = 0$. The lower eigenvalue is

$$\boxed{E_- = \frac{\Delta - \sqrt{\Delta^2 + 4v^2}}{2}.}$$

A normalized lower eigenstate is

$$\boxed{|\psi_-\rangle = \cos\vartheta|0\rangle - \sin\vartheta|d\rangle, \qquad \tan 2\vartheta = \frac{2v}{\Delta}, \quad 0 < \vartheta < \frac{\pi}{4}.}$$

For $v/\Delta \ll 1$,

$$\vartheta = \frac{v}{\Delta} + O\left(\frac{v^3}{\Delta^3}\right), \quad E_- = -\frac{v^2}{\Delta} + O\left(\frac{v^4}{\Delta^3}\right),$$

$$|\langle d|\psi_-\rangle|^2 = \frac{v^2}{\Delta^2} + O\left(\frac{v^4}{\Delta^4}\right).$$

Thus a nonzero source may mainly dress the reference state rather than create a continually growing excitation population.

For evolution from $|0\rangle$, subtract the irrelevant term $(\Delta/2)\mathbb{1}$:

$$H_X - \frac{\Delta}{2}\mathbb{1} = v\sigma_x - \frac{\Delta}{2}\sigma_z.$$

Let $\omega_X = \sqrt{\Delta^2 + 4v^2}$. Exponentiating this Pauli matrix gives

$$\boxed{P_d(t) = \frac{4v^2}{\Delta^2 + 4v^2} \sin^2\left(\frac{\sqrt{\Delta^2 + 4v^2}\, t}{2}\right).}$$

Therefore

$$P_d(t) = v^2 t^2 + O(t^4).$$

The exact probability oscillates and can return to zero. The initial curvature v^2 is not an irreversible decay rate.

Marking: Eigenvalue and state: 0.65 (C). Weak-mixing forms: 0.35 (C). Exact probability: 0.6 (C). Rate distinction: 0.2 (M).

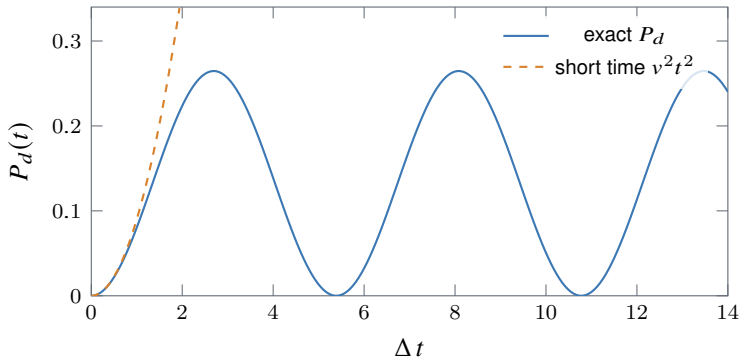


Figure 2.2. B.1 — oscillation, not decay (illustrative $v/\Delta = 0.3$). The exact probability oscillates between 0 and $4v^2/(\Delta^2 + 4v^2)$ and returns to zero. The short-time curvature v^2 describes only the start; it is not an irreversible decay rate.

B.2 Local improvement and new cross terms **0.8 pt**

For the overlap diagnostic, commute only the operators on spin 2:

$$[h_{23}, S_{12}] = -i\gamma\epsilon\, \sigma_1^x [\sigma_2^z, \sigma_2^x]\sigma_3^z$$
$$= \boxed{2\gamma\epsilon\, \sigma_1^x \sigma_2^y \sigma_3^z.}$$

This term acts on three sites. It was not present in the isolated 12-pair calculation. In fact, for real γ, ϵ its operator norm is $2|\gamma\epsilon|$.

A chain transformation combines overlapping corrections. Its transformed Hamiltonian contains cross commutators as well as the residual of each isolated cluster. It may also retain terms that change the reference configuration. Neither contribution can be discarded because the single-cluster residual decreased.

Marking: Commutator: 0.55 (C). Why single-cluster success does not certify the chain: 0.25 (M).

Part C. Exact frames and kept dynamics (3.1 points)

C.1 Keep the Hamiltonian, the frame velocity, and the initial state 1.8 pt

(a) Differentiate $|\zeta\rangle = \mathcal{U}^\dagger|\Psi\rangle$ and use unitarity:

$$i\partial_t|\zeta\rangle = \mathcal{K}_{\rm corr}|\zeta\rangle, \qquad \mathcal{K}_{\rm corr} = \mathcal{U}^\dagger H_{\rm MFP}\mathcal{U} - i\mathcal{U}^\dagger\dot{\mathcal{U}}.$$

Substitute $\mathcal{U} = We^S$:

$$\boxed{\mathcal{K}_{\rm corr} = e^{-S}\mathcal{K}_W e^S - i\int_0^1 e^{-uS}\dot{S}e^{uS}\,\mathrm{d}u.}$$

This is exact for any specified differentiable unitary frame and anti-Hermitian S.

(b) Expansion gives

$$\begin{aligned}\mathcal{K}_{\rm corr} = \mathcal{K}_W + [\mathcal{K}_W, S] - i\dot{S} \\ + \tfrac{1}{2}[[\mathcal{K}_W, S], S] - \tfrac{i}{2}[\dot{S}, S] + \cdots .\end{aligned}$$

The first line contains all first-order terms in the stated expansion. The frame's velocity contributes even when a static transformation reduces a residual at one phase. A static result alone does not bound $\dot{S}$ throughout a cycle.

(c) The equality $\mathcal{K}_{\rm corr} = \mathcal{K}_{\rm ret} + \mathcal{R}$ is only a partition. An approximation occurs when $\mathcal{R}$ is omitted or replaced. The exact physical initial density matrix becomes

$$\rho'(0) = \mathcal{U}^\dagger(0)\rho(0)\mathcal{U}(0).$$

Choosing a projected or simpler reference state instead adds an initial-state error. It is not a consequence of the basis transformation. A useful reduced calculation must bound the effect of $\mathcal{R}$, not just rename it.

(d) A one-particle rotation replaces each c_j by a linear combination of modes. A quartic interaction remains a sum of quartic terms, generally with more orbital indices. No high band has been removed by this operation. The displayed e^S specifies the form of a possible many-body dressing, not its values or its spatial decay.

Marking: (a) Exact generator: 0.6 (C). (b) First-order terms: 0.3 (C); the role of $\dot{S}$: 0.2 (M). (c) Where the approximation enters, transformed initial state, need for a bound: 0.4 (M). (d) Interaction stays two-body; e^S unspecified: 0.3 (M).

C.2 The retained dynamics can change the state 1.3 pt

(a) The fixed direction is preserved up to a phase if and only if, for every time in the interval,

$$\boxed{(1 - \Pi_A)\mathcal{K}_{\rm ret}(t)|A_0\rangle = 0.}$$

Equivalently $\mathcal{K}_{\rm ret}(t)|A_0\rangle = E_A(t)|A_0\rangle$. Then its solution is $e^{-i\int E_A\,dt}|A_0\rangle$. This statement concerns a fixed direction in the chosen frame; a moving reference needs its frame-velocity term.

(b) In the periodic teaching chain, the diagonal part gives $E_A = -NJ_z + Nh$. The transverse field gives

$$\boxed{(H_{\rm d} - E_A)|A_0\rangle = -g\sum_{j=1}^{N}|j\rangle.}$$

The $|j\rangle$ are mutually orthogonal, hence

$$\|(H_{\rm d} - E_A)|A_0\rangle\|^2 = Ng^2.$$

For a given cell, the first-order amplitude of its single flip is igt, and

$$\boxed{\langle q_j(t)\rangle = g^2t^2 + O(t^3).}$$

The global source norm grows with N, while this local initial coefficient does not. This is a short-time expansion, not a uniform long-time estimate.

(c) By construction, $\mathcal{K}_{\rm ret}$ is the exact Hamiltonian of this teaching example and $\mathcal{R} = 0$. There is no truncation error, yet the retained dynamics changes the reference direction. Model accuracy and reference-pump stability are different questions. No physical MFP current has been specified or evaluated here, so there is no MFP lifetime result.

Marking: (a) Invariance condition: 0.3 (M). (b) Source vector 0.3 (C), norm Ng^2 0.15 (C), $\langle q_j\rangle = g^2t^2$ 0.3 (C). (c) The three distinctions: 0.25 (M).

Part D. The physical current (3.0 points)

D.1 The same probability can give opposite currents 0.9 pt

Let $a = \sqrt{1-p}$ and $b = e^{i\beta}\sqrt{p}$. Then

$$\begin{aligned}\langle I\rangle &= -i\lambda\, a^* b + i\lambda\, b^* a \\ &= \boxed{2\lambda\sqrt{p(1-p)}\sin\beta.}\end{aligned}$$

Thus the three requested phases give

β	0	$\pi/2$	$-\pi/2$
$\langle I\rangle$	0	$2\lambda\sqrt{p(1-p)}$	$-2\lambda\sqrt{p(1-p)}$

For small p and a fixed nonzero $\sin\beta$, the leading magnitude is proportional to $\sqrt{p}$. A probability contains no information about the relative phase β and does not determine the current.

Marking: Expectation value: 0.55 (C). Phase comparison and scaling: 0.2 (C). Probability alone is insufficient: 0.15 (M).

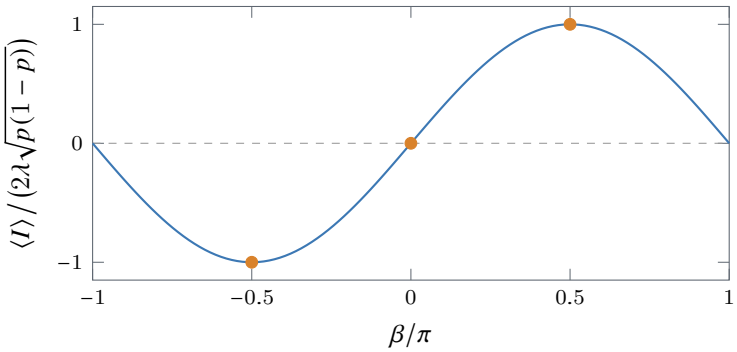


Figure 2.3. D.1 — the current follows the phase. $\langle I\rangle = 2\lambda\sqrt{p(1-p)}\sin\beta$. The same p gives zero, positive or negative current depending on β; at small p the size scales as $\sqrt{p}$.

D.2 A frame-dependent Hamiltonian derivative is incomplete 2.1 pt

Write $H' = \mathcal{U}^\dagger H \mathcal{U}$. From unitarity,

$$\partial_\theta H' = \mathcal{U}^\dagger(\partial_\theta H)\mathcal{U} + i[\mathcal{A}_\theta, H'].$$

Since $I = -\partial_\theta H$,

$$I' = -\partial_\theta H' + i[\mathcal{A}_\theta, H'].$$

Substitute $H' = \mathcal{K} + \mathcal{A}_t$:

$$I' = -\partial_\theta \mathcal{K} - \partial_\theta \mathcal{A}_t + i[\mathcal{A}_\theta, \mathcal{K}] + i[\mathcal{A}_\theta, \mathcal{A}_t].$$

Use the supplied connection identity to cancel the last commutator. The result is

$$\boxed{I' = -\partial_\theta \mathcal{K} - \partial_t \mathcal{A}_\theta - i[\mathcal{K}, \mathcal{A}_\theta].}$$

For $i\partial_t|\Xi\rangle = \mathcal{K}|\Xi\rangle$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \mathcal{A}_\theta \rangle = \langle \partial_t \mathcal{A}_\theta + i[\mathcal{K}, \mathcal{A}_\theta] \rangle \,.$$

Integrating the physical current therefore gives

$$\boxed{Q(t_1, t_2) = -\int_{t_1}^{t_2} \langle \partial_\theta \mathcal{K} \rangle \, \mathrm{d}t - \left[\langle \mathcal{A}_\theta \rangle \right]_{t_1}^{t_2}.}$$

All expectation values here use the same transformed physical state. If more than one gauge patch is used, patch seams must be included consistently.

A closed projector path fixes a subspace, not a unique frame or its holonomy. It does not imply that the endpoint term vanishes. The two terms are frame-dependent; their sum is the physical charge. In particular, the endpoint term alone is not automatically a Chern charge.

Marking: Differentiate H' and obtain the operator identity: 1.2 (C). Integrate using the equation of motion: 0.6 (C). Endpoint and gauge interpretation: 0.3 (M).

Problem 3: How long can the conveyor operate? (10.0 points)

Part A. Local accuracy in a large system (5.3 points)

A.1 A product-state diagnostic 1.1 pt

The global overlap amplitude is $(\sqrt{1-p})^N$, so

$$\boxed{\mathcal{F}_N = (1-p)^N.}$$

The reduced state of m cells is still the prescribed product state:

$$\boxed{\mathcal{F}_m = (1-p)^m.}$$

For fixed $p > 0$, $\mathcal{F}_N \to 0$ as $N \to \infty$, but $\mathcal{F}_m$ is independent of the rest of the chain. Poor global overlap alone cannot diagnose a large error in a fixed local observable. It also does not prove local transport stability.

Marking: Global fidelity: 0.45 (C). Local fidelity: 0.4 (C). Correct interpretation: 0.25 (M).

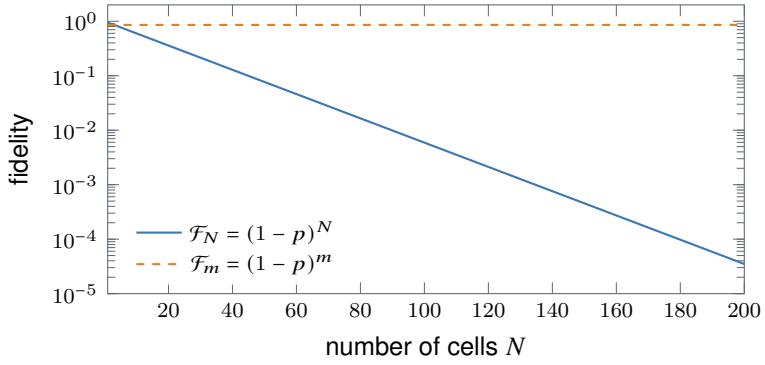


Figure 3.1. A.1 — global versus local fidelity (illustrative $p = 0.05$, $m = 3$). The global fidelity decays exponentially with N, while the fidelity of a fixed window does not change. A tiny global overlap says nothing, by itself, about a local current.

A.2 A trace-distance bound for the current 2.2 pt

Write

$$\Delta I = \mathrm{Tr}[(\rho' - \sigma')I_X] + \mathrm{Tr}[(\rho' - \sigma')(I' - I_X)].$$

The first term depends only on the states reduced to X. Apply the supplied norm inequalities:

$$\boxed{|\Delta I| \le \|I_X\| \left\|\rho'_X - \sigma'_X\right\|_1 + 2\eta_X.}$$

For a product reference, define $P_X = |A_{0,X}\rangle\langle A_{0,X}| = \prod_{x\in X}(1-q_x)$. Since $1 - P_X \le \sum_x q_x$,

$$1 - \mathrm{Tr}(\rho'_X P_X) \le D_X.$$

The pure-state trace-distance inequality, together with its maximum value 2, gives

$$\left\|\rho'_X - P_X\right\|_1 \le 2\min(1, \sqrt{D_X}).$$

Hence

$$\boxed{|\Delta I| \le 2\,\|I_X\|\min(1, \sqrt{D_X}) + 2\eta_X.}$$

For a general correlated reference, use the first trace-distance bound. Do not impose the product-reference expression for D_X without a justified frame and factorization.

Marking: General current bound: 1.1 (C). Product-reference bound: 0.85 (C). Scope and interpretation: 0.25 (M).

A.3 The boundary correlations carry dynamical information **2.0 pt**

(a) Take the partial trace of the reference equation. Since $\mathcal{K}_X$ acts inside X,

$$\mathrm{Tr}_{\bar{X}}[\mathcal{K}_X, \sigma'] = [\mathcal{K}_X, \sigma'_X].$$

Cyclicity for exterior operators gives $\mathrm{Tr}_{\bar{X}}[\mathcal{K}_{\bar{X}}, \sigma'] = 0$. Thus

$$\boxed{i\dot{\sigma}'_X = [\mathcal{K}_X, \sigma'_X] + \mathrm{Tr}_{\bar{X}}[\mathcal{K}_{\partial X}, \sigma'].}$$

(b) The last term generally depends on the state across the boundary, not on σ'_X alone. Even a short-range boundary Hamiltonian needs correlations with the nearby exterior. Under evolution those correlations can depend on farther degrees of freedom. A larger cluster, a controlled approximation, or a suitable error estimate is needed. The composite single-particle gap controls a different object; it does not specify this many-body correlation term.

(c) Subtract the two full equations:

$$\boxed{i\partial_t \delta\rho = [\mathcal{K}_{\mathrm{ret}}, \delta\rho] + [\mathcal{R}, \rho'].}$$

Its initial value is

$$\delta\rho(0) = \mathcal{U}^\dagger(0)\rho(0)\mathcal{U}(0) - \sigma'(0).$$

The first term propagates existing differences. The second contains the effect of omitted dynamics on the actual state. The identity is exact, but ρ' is not thereby known. Taking a partial trace does not remove the need for correlation information or a controlled estimate of this term.

Marking: (a) Local equation, with the exterior term vanishing: 0.75 (C). (b) Boundary correlations needed; the band gap does not supply them: 0.55 (M). (c) Difference equation and initial value: 0.45 (C); not yet a bound: 0.25 (M).

Part B. A conditional operating-time certificate **(1.6 points)**

B.1 Integrating the supplied local bounds **1.6 pt**

The given inequality includes $s = 0$, hence requires $D_X(0) = 0$. This is a classroom premise. Under the full stated hypothesis, $\min(1, \sqrt{D_X(s)}) \le \alpha s$ for $s \ge 0$, and A.2 gives

$$|\Delta I(s)| \le 2I_0\alpha s + 2\eta.$$

Integrate over a two-period window:

$$\begin{aligned} |\overline{Q} - \overline{Q}_{\mathrm{ref}}| &\le \frac{1}{2}\int_{t-2T}^{t} (2I_0\alpha s + 2\eta)\,\mathrm{d}s \\ &= 2I_0\alpha T(t-T) + 2T\eta. \end{aligned}$$

Add the supplied reference-charge error:

$$\boxed{|\overline{Q}(t) - Q_*| \le \delta_0 + 2I_0\alpha T(t-T) + 2T\eta.}$$

For $\alpha I_0 > 0$, an algebraic endpoint is

$$\boxed{t_{\mathrm{cert}} = T + \frac{\delta - \delta_0 - 2T\eta}{2I_0\alpha T}.}$$

Require $\delta - \delta_0 - 2T\eta > 0$. The useful certified interval is

$$2T \le t \le \min(t_{\max}, t_{\mathrm{cert}}),$$

and it is nonempty only if this upper endpoint is at least $2T$.

For the numerical exercise, express time in $\tau_0 = T$. Then

$$\frac{t_{\mathrm{cert}}}{T} = 1 + \frac{0.005 - 0.001 - 0.00002}{0.00002} = \boxed{200}.$$

The assumptions were stipulated to hold until $201T$. Thus all stated windows ending from $2T$ through $200T$ are certified. The next period need not fail: the bound has merely stopped certifying it.

Marking: Integrated bound: 0.55 (C). Endpoint and conditions: 0.45 (C). Numerical value: 0.35 (C). Lower-bound interpretation: 0.25 (M).

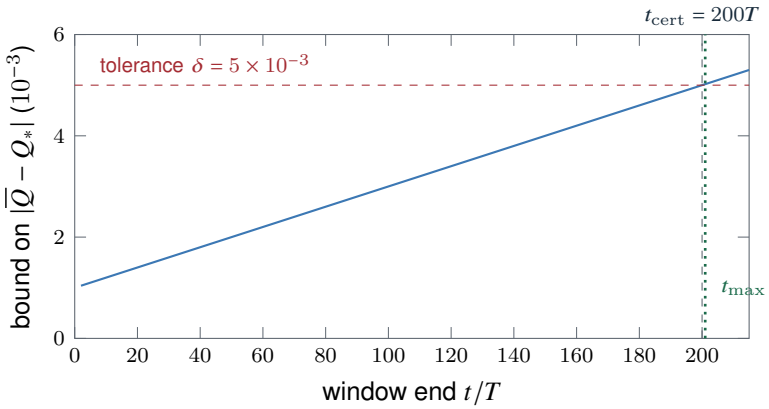


Figure 3.2. B.1 — where the certificate ends. The bound $\delta_0 + 2T\eta + 2I_0\alpha T(t - T)$ grows linearly and reaches δ at $t = 200T$. After that the bound no longer certifies anything; it does not predict a failure.

Part C. Prediction, error and first failure (3.1 points)

C.1 Use intervals, not an unqualified platform label 1.7 pt

(a) Integration gives the computable candidate budget

$$\boxed{E(t) = \frac{1}{2}\int_{t-2T}^{t} \left[b_X(s)e_X(s) + 2\eta(s)\right] \mathrm{d}s.}$$

Then $|\overline{Q} - \widehat{Q}| \le E$ if all supplied bounds hold on the window. The local-state estimate e_X must include any initial mismatch. If the reference charge is evaluated with a truncated operator or an approximate integrator, its additional error must also be included; the definition here used the full I'.

(b) The triangle inequality and its reverse give

$$|\widehat{Q} - Q_*| + E \le \delta \quad \Rightarrow \text{certified inside},$$
$$|\widehat{Q} - Q_*| - E > \delta \quad \Rightarrow \text{certified outside}.$$

If neither holds, the classification is undetermined. The second inequality is strict because equality still satisfies the specified tolerance.

(c) For the supplied arithmetic,

Window	$\lvert\widehat{Q} - Q_*\rvert + E$	$\lvert\widehat{Q} - Q_*\rvert - E$	Decision
1	0.0040	0.0020	Inside
2	0.0075	0.0045	Undetermined
3	0.0065	0.0055	Outside

The first failed window among these three samples may be 2 or 3. The data do not decide which. Windows at unlisted start times are uncontrolled, so the first continuous-start failure is not determined. A different persistence rule would require additional consecutive-window information.

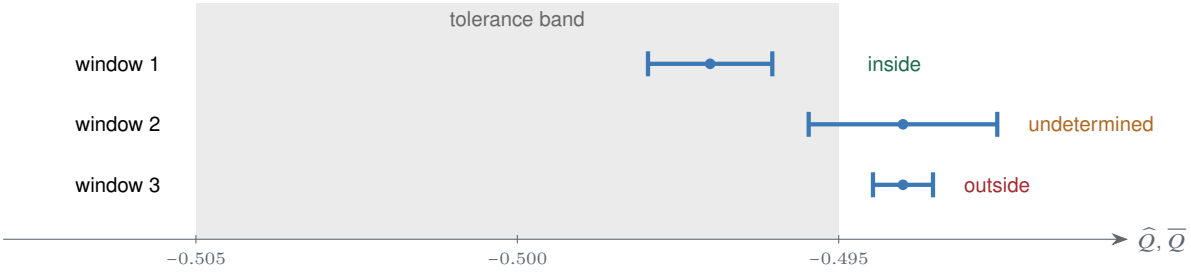


Figure 3.3. D.1 — three sample windows. Each bar is $\widehat{Q} \pm E$. Window 1 lies wholly inside the band, window 3 wholly outside, and window 2 straddles the edge, so it is undetermined. Whether window 2 or 3 is the first failure cannot be decided from these data.

Marking: (a) $E(t)$: 0.4 (C); where the initial mismatch enters: 0.25 (M). (b) Both tests and the undetermined case: 0.6 (M). (c) Classification: 0.25 (C); undetermined first failure: 0.2 (M).

C.2 The missing inputs are the research **1.4 pt**

A candidate description has the structure

$$|\Psi_{\mathrm{ref}}(t)\rangle = W(t)e^{S(t)}|\Xi(t)\rangle.$$

Here W follows a geometric reference frame, S supplies correlation corrections, and $|\Xi\rangle$ retains physical configuration mixing. If the retained Hamiltonian changes the reference direction, $|\Xi\rangle$ cannot be replaced by one overall phase.

What must be supplied by the research model

1. A reference defined without using the unknown exact long-time state, with controlled locality, time dependence, and two-period seams.
2. A description of retained correlated dynamics and of all omitted effects. Higher-band terms and overlapping corrections must be included or bounded.
3. A physical-current prediction, with the correct transformed operator, initial state, and boundary terms.
4. A useful local error estimate over complete cycles. It must distinguish an upper bound from a fitted error curve.

If the final result has the form

$$|\overline{Q}(t) - \widehat{Q}(t)| \le E(t),$$

then the triangle inequality and reverse triangle inequality give

$$|\widehat{Q} - Q_*| + E \le 0.005 \quad \Longrightarrow \text{window certified to pass,}$$
$$|\widehat{Q} - Q_*| - E > 0.005 \quad \Longrightarrow \text{window certified to fail.}$$

Neither implication, by itself, locates the first failure. Earlier windows, any persistence rule, and gaps between sampled start times need control.

Marking: Reference with controlled seams: 0.25 (M). Kept dynamics and omitted effects: 0.25 (M). Physical-current prediction: 0.25 (M). Useful local error over full cycles: 0.2 (M). First failure needs earlier windows, any persistence rule and gaps between starts: 0.45 (M).

Appendices: research context

Appendices A–C are for teachers and for readers who want the research picture behind the exam; they are not needed to solve or mark any question. Appendix D, the toolbox, is for the reader. Labels have the form problem.part.number.

Appendix A. Paper map

What each paper answers

- **Paper [1].** Question: can repulsion turn an integer Thouless pump into one that carries half a particle per period at integer filling, and how? Answer: yes, at strong repulsion in small systems: the particles occupy one of two localized two-band orbital types, which exchange every period, so each particle moves one cell in two periods. In large systems this should last only for a finite (prethermal) time, which the paper leaves open. Problems 1 and 2 lead to this mechanism; Problem 3 is built on the open question.
- **Paper [2].** Question: how can a moving group of bands be described without choosing a basis inside it? Answer: by the band projector, its derivative (a tangent vector) and the generator built from them; metric, curvature, holonomy and topology follow from these objects. Problem 1, Parts B–D use this language.

Research target: predict when the original interacting pump leaves a chosen charge tolerance, and identify the process that controls that change. A guaranteed interval is useful, but is not automatically the actual first-failure time.

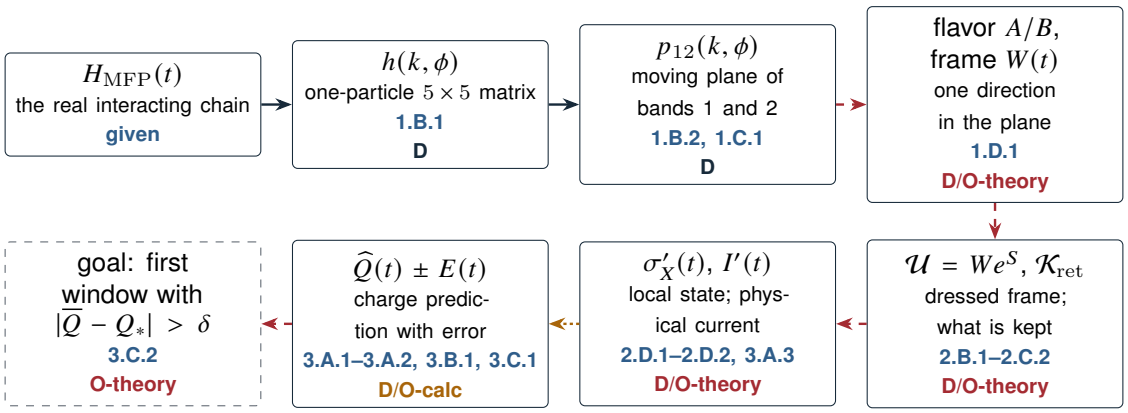


Figure A.1. The chain of objects behind the three problems. Each box is a mathematical object; the blue labels give the questions (problem.part.number) that work with it, and the last line gives the kind of the step that leads to it. Black arrows (D): derived in the exam, nothing open. Amber dotted arrows (O-calc): open, but existing theory is enough and a calculation is missing. Red dashed arrows (O-theory): open, because the closest existing theory needs a hypothesis that fails here. D/O: the identity is derived in the exam; its inputs for the original pump are open.

Language map

Each term of the papers in three forms: the physical process or picture it names, the familiar equivalent a physicist from another field already knows, and how it is computed or measured. “New” marks a concept with no familiar equivalent; the listed questions teach it.

Paper term	Physical process or picture	Familiar equivalent	How it is computed or measured
Thouless pump	a slow periodic change of the lattice that carries particles along it, cycle after cycle	adiabatic transport; Berry phase	charge through one bond per period: time integral of the bond current
Chern number of a band	the integer number of cells the charge of a filled band moves per period	quantized Hall conductance	winding of the Wannier centres over one period (Fig. 1.3a)
Wannier orbital, centre	a localized orbital built from the Bloch states of a band; where it sits	tight-binding orbital; Fourier series	$\langle w\|x\|w\rangle$; eigenphases of the projected position operator
flavor A/B	two localized orbital types per cell built from bands 1 and 2; a particle occupies one and rides with it	basis choice inside a degenerate subspace	eigenvectors of the position operator projected onto bands 1 and 2
two-cycle flavor exchange	an A orbital slides into a B position each period and is back on A only after two periods	new	taught by 1.A.1 and 1.D.1; Fig. 1.3b
composite bands, $\mathrm{Gr}(2,5)$	bands 1 and 2 treated as one unit; the plane they span at each (k,ϕ)	projector onto a degenerate subspace	$p_{12}=\sum_{n=1,2}\|u_n\rangle\langle u_n\|$ (1.B.1)
horizontal generator	the rotation that carries the plane along its path without spinning it inside	Kato’s adiabatic generator; rotating frame	$g_t = i[\dot{p},p]$ (1.B.2, 1.C.1)
Wilson loop	going once around the Brillouin zone while carrying the band basis along	Berry phase on a closed path	ordered product of band-overlap matrices; eigenphases give the centres (1.D.1)
dressing e^S	building the main correlations into the reference state by a change of basis	Schrieffer–Wolff transformation	$e^{-S}He^S$ expanded in commutators (2.B.1, 2.C.1)
prethermal lifetime	how long a driven system stays in a long-lived non-equilibrium state before domains destroy the order	metastable state; false-vacuum decay	first time the transport leaves a tolerance; open in [1] (Problem 3)

Open links: a calculation or a new theory?

An open link is O-calc when a procedure with existing methods would close it and the hypotheses of those methods hold for the original pump. It is O-theory when the closest existing theory needs a hypothesis that fails here. The label says whether a method exists, not how much work remains.

Open link (questions)	Kind	Closest existing theory	Hypothesis that fails, or the calculation that closes it
the particles keep following one flavor (1.A.1, 1.D.1)	D/O-theory	many-body adiabatic theorem; many-body Chern number (Niu–Thouless–Wu)	a gap above the two flavor configurations, uniform in system size and over the cycle; the composite gap protects only the plane, not the direction in it
local dressing assembled along the chain and the cycle (2.B.1, 2.B.2)	O-calc	Schrieffer–Wolff expansion; locality (cluster) estimates	build the corrections at every phase, join them smoothly, bound the overlap terms
the kept dynamics $\mathcal{K}_{\mathrm{ret}}$ (2.C.1, 2.C.2)	D/O-theory	quasi-adiabatic continuation (Hastings); adiabatic theorems for extended systems (Bachmann–De Roeck–Fraas; Monaco–Teufel)	a spectral gap uniform in system size
the local state over the full drive (3.A.3)	D/O-theory	Lieb–Robinson and local perturbation bounds	the bounds grow with time; the boundary correlations do not close at the times of interest
the certificate inputs $\alpha, I_0, \eta, \delta_0, t_{\max}$ (3.B.1)	D/O-calc	trace-distance and locality bounds	evaluate them in one frame for one initial state, once the local state is controlled
the first failed window (3.C.2)	O-theory	rigorous prethermalization (Abanin–De Roeck–Ho–Huveneers; Else–Bauer–Nayak)	a fast drive or a large gap; here the drive is slow

NOT ESTABLISHED FOR MB1 — OPEN — the shared gap

The O-theory links fail on the same point: no existing theory controls a slowly driven, strongly correlated chain over long times when its order is not protected by a uniform gap. A first controlled result is a limit in which the failing hypothesis holds, for example a gap that protects the whole group of flavor configurations rather than one of them.

Appendix B. Source, scope and open conditions for each question

Green frames name the source of a result and its scope; red frames mark teaching models, assumed bounds and open conditions. These notes were removed from the question paper so that they do not interfere with solving.

Problem 1: A two-cycle quantum conveyor: bands, projectors and flavors

1.A.1

NOT ESTABLISHED FOR MB1 — OPEN — the central dynamical link

The reference evolution and its local error are not yet controlled over the full drive. In particular, $\mathcal{K}_{\mathrm{ret}} \to \sigma'_X \to \widehat{Q}$ is not supplied by the composite-band geometry.

AVAILABLE — MFP [1] — a candidate two-cycle branch

The two Wannier flavors may exchange after one period. In the source argument, a continuously followed flavor returns after two periods with a one-cell displacement (introduction of Problem 1 and Figure 1.3 of the question paper). The argument assumes that the occupations continue to follow that branch.

NOT ESTABLISHED FOR MB1 — ADDED IDEAL-FOLLOWING CONDITION

For 1.A.1, stipulate that the actual occupied orbitals follow the branch exactly. This is not a large-system finite-speed stability result.
An N-cell ring has cell length a and contains N particles. After $2T$, each occupied orbital has an unwrapped center displacement $\Delta x_j = -a$. Define the spatially averaged transported particle number by

$$Q(0, 2T) = \frac{1}{Na} \sum_{j=1}^{N} \Delta x_j .$$

The single-period values $Q(0, T)$ and $Q(T, 2T)$ are defined in the same way from the displacements during each period.

AVAILABLE — MFP [1] — what the source supplies

The source's branch-following argument connects two-period flavor winding to fractional average transport. 1.A.1 isolates this counting step.

NOT ESTABLISHED FOR MB1 — OPEN — tracking was stipulated

The argument has assumed actual following. It has not bounded excitation, internal mixing, or long-time transport error in a large interacting system.

1.B.1

AVAILABLE — MFP [1], Eq. (1)–(2) — source model and motivation

In the source mechanism, mixing the lowest two bands permits more localized Wannier orbitals, and finite-range repulsion can favor occupation of one flavor (introduction of Problem 1).

NOT ESTABLISHED FOR MB1 — OPEN — a reference choice is not a dynamical truncation

The two-band N-particle sector is not a two-state system. Restricting the full interacting model to this sector requires control of the terms that leave it.

Connection to the next step: The physical output is a defined composite-band projector. 1.B.2 and 1.C.1 describe its motion. 1.D.1 then asks how to identify a flavor inside it.

Connection to the next step: Before extracting geometry, name the projector, its ambient space, and the physical reason for using it. Do not identify p_{12} with a many-body low-energy pair.

1.B.2

AVAILABLE — H2 [2] — projector kinematics

A projector represents a subspace without choosing a basis inside it. Its derivative describes the change of that subspace. In 1.B.2, derive the corresponding horizontal generator. This is a kinematic statement, not a lifetime prediction.
Let $p(t)$ be a smooth orthogonal projector, so $p^2 = p = p^\dagger$.

AVAILABLE — H2 [2] — exact kinematics

The horizontal-generator construction applies to the given smooth projector. It does not supply the missing many-body dynamics or a transition rate.

1.C.1

NOT ESTABLISHED FOR MB1 — ADDED MODEL — an exactly solvable rotating spin

For 1.C.1, study one spin. This is a test of moving-frame dynamics, not the complete interacting pump.

$$H(t) = -\frac{\Delta}{2}\big[\sin(\nu t)\sigma_x + \cos(\nu t)\sigma_z\big], \qquad \Delta, \nu > 0.$$

Its instantaneous ground-state projector and a rotating frame are

$$p(t) = \tfrac{1}{2}\big[\mathbb{1} + \sin(\nu t)\sigma_x + \cos(\nu t)\sigma_z\big],$$
$$W(t) = e^{-i\nu t\sigma_y/2}, \qquad |\Xi(t)\rangle = W^\dagger(t)|\psi(t)\rangle.$$

Use $|\psi(0)\rangle = |\uparrow\rangle$, where $\sigma_z|\uparrow\rangle = |\uparrow\rangle$.

Source / scope: 1.B.2 uses H2. 1.C.1 is a teaching construction. Its exact solution does not prove a regime of the original pump.

1.D.1

NOT ESTABLISHED FOR MB1 — OPEN — flavor and correlated-frame conditions

p_{12} is a composite projector, not a selected flavor projector. It does not fix the internal unitary of the two-band group, and its smoothness does not provide a full interacting reference state.

NOT ESTABLISHED FOR MB1 — ADDED COUNTEREXAMPLE — not computed MFP matrices

In a physically labelled two-dimensional space, compare $\mathcal{T}_0 = \mathbb{1}_2$ with $\mathcal{T}_1 = i\sigma_x$. Both act inside the same plane. No real-space displacement is specified. These are illustrative internal maps, not the original pump's Wilson or evolution matrices.

AVAILABLE — MFP [1], Eq. (2); H2 [2] — two linked objects

The source chooses localized Wannier flavors inside the composite bands. H2 describes the composite projector without an arbitrary internal basis. A physical flavor requires a localization rule and continuous tracking, not just the endpoint statement $p_{12}(T) = p_{12}(0)$.

AVAILABLE — SOURCE CONNECTION — where H2 enters

MFP supplies a physical composite-band and Wannier construction. H2 organizes subspace motion and its internal-basis freedom. The bridge makes their objects explicit; it does not invent a new dynamical law.

NOT ESTABLISHED FOR MB1 — OPEN — not an automatic rank-one upgrade

A gap and locality bounds for the composite bands would not certify the selected flavor's full internal rotation or seam. Nor would they prove that an interacting state occupies that branch over the drive. These are unchanged research conditions.

Connection to the next step: Keep the full interactions when using the geometric frame. 2.C.1 explains this step exactly.

2.A.1

NOT ESTABLISHED FOR MB1 — ADDED MODEL — a spin-chain analogy

For 2.A.1 only, replace the pump by the model below. Its parameters are not identified with K or U_r. This model explains an energy competition; it is not a controlled reduction of MFP.

$$H_{\mathrm{d}} = -J_z \sum_j \sigma_j^z \sigma_{j+1}^z - h \sum_j \sigma_j^z - g \sum_j \sigma_j^x, \qquad J_z, h > 0.$$

Let A mean $\sigma^z = -1$ and B mean $\sigma^z = +1$. Set $g = 0$. Start with all spins in A. Flip a consecutive segment of ℓ spins to B, with $1 \le \ell < N$. There are two interfaces between the segment and its surroundings.

NOT ESTABLISHED FOR MB1 — ADDED MODEL — scope of the answer

This exact answer belongs to the teaching spin model. Deriving the domain variables, couplings, and driven transitions from MFP is a separate task.

2.B.1

NOT ESTABLISHED FOR MB1 — ADDED MODEL — one double-excitation channel

For 2.B.1, retain only a bare two-cell reference $|0\rangle = |o, o\rangle$ (both particles in the reference orbital o) and one double excitation $|d\rangle$. A real local calculation has many more channels; one denominator cannot replace that spectrum without a separate approximation.

NOT ESTABLISHED FOR MB1 — ADDED MODEL — one channel is not the full local spectrum

The exact two-channel solution is not a fit of the full local response. A controlled reduction would need the actual channel energies, matrix elements, retained processes, and errors. The gap Δ here is a teaching parameter.

2.B.2

Source / scope: Use $[\sigma_z, \sigma_x] = 2i\sigma_y$. Operators on different spins commute.

NOT ESTABLISHED FOR MB1 — OPEN — compatibility along the chain and drive

A static correction at one phase does not provide a smooth full-cycle correction, its time derivative, or its two-cycle seam. Smaller isolated-cluster norms do not certify a small global or local interaction remainder after assembly.

Teacher's distinction

There are three separate statements:

1. A particular pair has a more accurate correlated reference.
2. The pair corrections can be combined into a controlled chain description.
3. That description predicts the physical current during the full drive.

A calculation on isolated pairs can support the first statement. The second and third need separate arguments; 2.B.2 shows the first obstacle to the second.

2.C.1

Connection to the next step: 2.C.2 tests whether the kept part actually preserves the desired reference.

NOT ESTABLISHED FOR MB1 — OPEN — a useful retained model

One must still choose and solve $\mathcal{K}_{\mathrm{ret}}$, control the omitted terms and observables, and retain any physical mixing that changes the current. A small model error does not imply a perfectly quantized reference pump.

2.C.2

NOT ESTABLISHED FOR MB1 — ADDED MODEL — reuse 2.A.1, do not infer an MB1 mapping

The spin chain in (b) is still the teaching analogy, not the full MFP Hamiltonian. Use a ring of N spins. Set $E_A = \langle A_0|H_{\mathrm{d}}|A_0\rangle$. Let $|j\rangle$ denote the orthonormal configuration with only spin j flipped. You may use $e^{-iHt} = \mathbb{1} - iHt + O(t^2)$.

NOT ESTABLISHED FOR MB1 — ADDED MODEL — what the exact answer does not establish

The spin example is only a counterexample to the claim that zero omitted remainder implies no physical evolution. Its g^2 coefficient cannot be inserted into 3.B.1 as an MB1 error-growth bound.

Connection to the next step: Predict $|\Xi\rangle$ or its relevant local reduced states. Do not first impose ideal charge on a reference whose retained dynamics has not been solved.

2.D.1

NOT ESTABLISHED FOR MB1 — ADDED MODEL — a current-connected pair of states

For 2.D.1, $|0\rangle$ and $|e\rangle$ are any orthogonal states connected by the stated current. Do not identify $|e\rangle$ with 2.B.1's pure double excitation without checking the current matrix element. A one-body hopping current does not directly connect Slater determinants differing by two occupied orbitals.

$$I = \lambda \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad |\psi\rangle = \sqrt{1-p}|0\rangle + e^{i\beta}\sqrt{p}|e\rangle, \quad \lambda \in \mathbb{R}, \quad 0 \le p \le 1.$$

NOT ESTABLISHED FOR MB1 — ADDED MODEL — the current matrix is part of the question

2.D.1 assumes a current-connected pair. The example isolates interference. It does not specify how the physical MFP evolution generates that pair or its phase.

Why the current-connected state is not automatically a double excitation

A hopping current has the one-body form

$$I = \sum_{ab} I_{ab} d_a^\dagger d_b.$$

It changes at most one occupied orbital of a Slater determinant. If $|\Phi_{ij}^{ab}\rangle$ differs from $|\Phi\rangle$ by two occupied orbitals, then

$$\langle \Phi_{ij}^{ab} | I | \Phi \rangle = 0.$$

A double-excitation source can still influence later dynamics and the current of a correlated state. The vanishing direct matrix element only forbids identifying every source norm with the two-state current amplitude λ.

2.D.2

NOT ESTABLISHED FOR MB1 — OPEN — solving the expectation values

The full research problem still needs the driven correlated state, the physical current in that state, and the correct reference charge. Rewriting the current is not yet predicting it.

3.A.1

NOT ESTABLISHED FOR MB1 — ADDED STATE — a diagnostic product family

For 3.A.1 only, prescribe the state below. It is not asserted to be a solution of the interacting pump. The scalar p is independent of N. Use the squared-fidelity convention $\mathcal{F}(\rho, |a\rangle) = \langle a|\rho|a\rangle$.

$$|\psi_N\rangle = \bigotimes_{j=1}^{N} \left[\sqrt{1-p}|A\rangle_j + \sqrt{p}|B\rangle_j\right], \qquad 0 < p < 1, \quad \langle A|B\rangle = 0.$$

3.A.2

Source / scope: Here $\|\cdot\|$ is the operator norm and $\|\cdot\|_1$ is the trace norm. The local q_x are reference-deviation operators, not automatically physical domain counters.

NOT ESTABLISHED FOR MB1 — OPEN — the unknown dynamical input

The inequality does not predict $\rho'_X - \sigma'_X$. Nor does it prove that D_X is a domain density or a decay rate. Establishing a useful full-cycle bound on these quantities remains part of the research.

3.A.3

Connection to the next step: 2.D.2 supplies the correct current operator. 3.A.2 converts a local-state error into a current error. 3.A.3 identifies the missing dynamics between these two tools.

NOT ESTABLISHED FOR MB1 — OPEN — preserve the initial state and the model

A nonzero initial difference cannot be set to zero by renormalizing a projected state. An assumed bath or factorization changes the approximation and needs its own justification. A local norm estimate must remain useful over the time interval and as the chain grows.

Connection to the next step: The project must compute a physical reference charge and a usable error, not only a geometric trajectory. 3.C.1 gives the conditional decision rule without assigning an ideal charge to the reference.

3.B.1

NOT ESTABLISHED FOR MB1 — GIVEN BOUNDS — the missing long-time inputs

Let $\alpha, I_0 > 0$ and $\eta, \delta_0 \geq 0$. Assume that, for every $0 \leq s \leq t_{\max}$,

$$D_X(s) \leq \alpha^2 s^2, \qquad \|I_X(s)\| \leq I_0, \qquad \eta_X(s) \leq \eta.$$

Also assume that every relevant two-period reference window satisfies $|\overline{Q}_{\mathrm{ref}}(t) - Q_*| \leq \delta_0$. These remain supplied assumptions for the full correlated MB1 frame. In particular, the bound at $s = 0$ requires $D_X(0) = 0$. Initial mismatch must be controlled separately; do not silently replace the physical initial state by its projection. Here α, I_0, and η have units of inverse time; δ_0 and charge are dimensionless.

NOT ESTABLISHED FOR MB1 — ILLUSTRATIVE ONLY — no MB1 lifetime has been computed

The value $200T$ follows from illustrative classroom inputs. The complete correlated-frame bounds, their common time interval, and the required initial matching have not been established for MB1.

Connection to 2.C.2

The product-reference form of 3.A.2 uses $\sigma'_X = |A_{0,X}\rangle\langle A_{0,X}|$ at every time s in the window. This is consistent only if the kept dynamics leaves the reference direction invariant, which is the 2.C.2(a) condition

$$(1 - \Pi_A)\,\mathcal{K}_{\mathrm{ret}}(s)|A_0\rangle = 0 \qquad (0 \leq s \leq t_{\max}),$$

together with the frame-velocity term if the reference direction moves. 2.C.2(b) gives a counterexample with $\mathcal{R} = 0$: the source $-g\sum_j |j\rangle$ moves the reference at order t. When the condition fails, D_X measured against a fixed $|A_{0,X}\rangle$ no longer measures $\rho'_X - \sigma'_X$. Use the general bound $\|I_X\|\left\|\rho'_X - \sigma'_X\right\|_1 + 2\eta_X$ with the evolving reference, and compute $\overline{Q}_{\mathrm{ref}}$ from that reference as in 3.C.1. A student who notes this restriction in 3.B.1 deserves credit under “conditions”.

3.C.1

Connection to the next step: 3.B.1 certifies an interval once special growth bounds are supplied. 3.C.1 permits a reference that itself changes or fails. Both require model-derived inputs before either gives an MB1 prediction.

AVAILABLE — RESEARCH LOGIC — preserve the final research goal

A useful reference predicts $\widehat{Q}(t)$; it need not remain ideal forever. Error control can then certify a stable interval or an actual failed window. Predicting first failure and its physical mechanism requires control of the preceding dynamics as well.

NOT ESTABLISHED FOR MB1 — GIVEN FUNCTIONS AND NUMBERS — teaching inputs only

The functions above are not established MB1 bounds. For the arithmetic test take $Q_* = -0.5$ and $\delta = 0.005$:

Ordered sample window	$\widehat{Q}$	E
1	-0.4970	0.0010
2	-0.4940	0.0015
3	-0.4940	0.0005

These are invented classification examples, not new trajectories. Use one-window exceedance; no persistence rule is imposed in this small exercise.

NOT ESTABLISHED FOR MB1 — ADDED DATA — no lifetime result

These decisions follow from supplied numerical error budgets. They are not evidence that an MFP simulation passed or failed at these times. The $200T$ example in 3.B.1 remains an independent classroom calculation.

3.C.2

NOT ESTABLISHED FOR MB1 — OPEN — the research task

Starting from $H_{\mathrm{MFP}}(t)$ and a specified physical initial state, construct a correlated reference evolution and a computable error estimate over complete drive cycles. The result must remain useful as the system size increases. Do not import 3.B.1's long-time assumptions as facts.

NOT ESTABLISHED FOR MB1 — The main scientific gap

The composite-band geometry does not supply a controlled full-cycle correlated reference, its physical current, or a quantitative large-system lifetime. Red frames mark those missing links.

NOT ESTABLISHED FOR MB1 — OPEN — no standard numerical answer is claimed

The scored exercise supplied several assumptions. The paper must derive suitable replacements from the microscopic MFP model and a specified physical initial state. The final task is not another substitution into the classroom formula.

NOT ESTABLISHED FOR MB1 — OPEN — lower bounds versus actual failure

A guaranteed operating interval and a prediction of actual first failure are different outputs. A small static source, a source norm, or a single favorable domain energy is not sufficient for either one without the missing dynamics and current analysis.

Appendix C. Teacher notes

What this example is

A public excerpt of a 26-question ResearchPhO paper. Questions that used unpublished calculations or proofs of the underlying research project were removed; the remaining 18 questions were relabelled within each problem and their points rescaled to 10 per problem, with every marking line rescaled in 0.05 steps.

The computed pictures and a sign observation

`figures.py` computes the band pictures from $h(k, \phi)$ and the Wannier centres from the projected position operator $Pe^{2\pi i x/L}P$ on a 24-cell ring at $K/J = -3/2$. With the Hamiltonian exactly as printed in [1], orbitals of band 1 move by $+2$ cells per period, and the followed composite flavor moves by -1 cell in two periods. An explicit time evolution of the filled band 1 on a 12-cell ring ($T = 400/J$) gives a rightward charge of $+2.00$ per period through every bond. These signs agree with the values $Q = +2$ and $Q = -1/2$ in Fig. 4 of [1] if Q is counted rightward-positive, as in this exam. For this Hamiltonian, the current expression printed in [1], $i\langle J_j c_j^\dagger c_{j+1} - J_j c_{j+1}^\dagger c_j\rangle$, is minus the rightward bond current, so [1] presumably uses a convention not stated there.

Sources

[1] M. Jürgensen, J. Steiner, G. Refael, and M. C. Rechtsman, *Multi-band fractional Thouless pumps*, arXiv:2504.09338v2 (2025). `arXiv:2504.09338`.

[2] S.-M. Huang, *Quantum-State Projectors on Grassmannian: Geometry, Holonomy, and Topology*, arXiv:2608.06777v1 (2026). `arXiv:2608.06777`.

[3] APhO 2025 theoretical question booklet, used as a layout reference only.

Appendix D. Toolbox: the smallest worked examples

One entry for each tool that the question paper states in "Useful information", in a hint or as a given identity, and that lies above an undergraduate who knows the four core physics courses. Each entry works the smallest case in which the tool does something. The examples teach the tools; none of them is an exam step. `check_answers.py` recomputes every result below.

A Bloch matrix with a seam (Problem 1, hint to B.1)

Statement. Insert $c_{R,s} = N^{-1/2}\sum_k e^{ikR}c_{k,s}$ into each term and use $\frac{1}{N}\sum_R e^{i(k'-k)R} = \delta_{kk'}$.

Smallest example. Take three sites per cell, the bonds t_0 $(0 \to 1)$ and t_1 $(1 \to 2)$ inside the cell, and the seam bond t_2 from orbital 2 of cell R to orbital 0 of cell $R+1$:

$$H = -\sum_R \left[t_0 c^\dagger_{R,1}c_{R,0} + t_1 c^\dagger_{R,2}c_{R,1} + t_2 c^\dagger_{R+1,0}c_{R,2} + \text{h.c.}\right].$$

Only the seam term changes the cell: $\sum_R c^\dagger_{R+1,0}c_{R,2} = \sum_k e^{-ik}c^\dagger_{k,0}c_{k,2}$. Hence

$$h(k) = -\begin{pmatrix} 0 & t_0 & t_2 e^{-ik} \\ t_0 & 0 & t_1 \\ t_2 e^{ik} & t_1 & 0 \end{pmatrix}, \qquad \det h(k) = -2t_0t_1t_2\cos k.$$

The momentum enters only through the product of the hoppings once around the cell. Three sites is the smallest case with a separate seam entry: with two sites per cell the seam bond and the inner bond join the same two orbitals, and their terms add in one entry, $-(t_0 + t_1 e^{-ik})$.

Used in 1.B.1(a).

Centres from the projected position operator (Problem 1, Useful information)

Statement. For a long ring of length L, the eigenvalues of $Pe^{2\pi i x/L}P$ on the range of P are close to $e^{2\pi i x_\alpha/L}$, where x_α are the centres of the localized orbitals.

Smallest example. A ring of N cells of length $a = 1$ $(L = N)$, two sites per cell at $x = R$ and $x = R + \frac{1}{2}$, joined by a hopping inside each cell only. The lower band consists of the bonding orbitals $|w_R\rangle = (|R,0\rangle + |R,1\rangle)/\sqrt{2}$, with centre $\langle w_R|x|w_R\rangle = R + \frac{1}{4}$. The operator $e^{2\pi i x/N}$ acts inside each cell, so $Pe^{2\pi i x/N}P = \sum_R \lambda_R |w_R\rangle\langle w_R|$ with

$$\lambda_R = \tfrac{1}{2}\left(e^{2\pi i R/N} + e^{2\pi i(R+1/2)/N}\right) = \cos\frac{\pi}{2N}\; e^{2\pi i(R+1/4)/N}.$$

The phase gives the centre $R + \frac{1}{4}$ exactly. The modulus $\cos(\pi/2N) < 1$ tends to 1 as N grows (0.924 at $N = 4$); this is the meaning of "close to".

Used in 1.D.1(b); Figure 1.3.

The Pauli exponential (Problem 1, Useful information)

Statement. For a real unit vector $\boldsymbol{d}$ and real x: $e^{ix\boldsymbol{d}\cdot\boldsymbol{\sigma}} = \mathbb{1}\cos x + i(\boldsymbol{d}\cdot\boldsymbol{\sigma})\sin x$.

Why. $(\boldsymbol{d}\cdot\boldsymbol{\sigma})^2 = \mathbb{1}$, so the even powers of the exponential series add up to $\mathbb{1}\cos x$ and the odd powers to $i(\boldsymbol{d}\cdot\boldsymbol{\sigma})\sin x$.

Smallest example. For $\boldsymbol{d} = \hat{z}$ the matrix is diagonal: $e^{ix\sigma_z} = \mathrm{diag}(e^{ix}, e^{-ix}) = \mathbb{1}\cos x + i\sigma_z \sin x$, which is Euler's formula twice. For $\boldsymbol{d} = \hat{x}$ and $x = \pi/2$: $e^{i\pi\sigma_x/2} = i\sigma_x$, which exchanges $|\uparrow\rangle$ and $|\downarrow\rangle$ up to the factor i. With $x = -\theta/2$, the operator $e^{-i\theta\,\boldsymbol{d}\cdot\boldsymbol{\sigma}/2}$ turns every Bloch vector by the angle θ about $\boldsymbol{d}$.

Used in 1.C.1, 1.D.1(c).

A change of frame (Problem 2, Useful information; also 1.C.1)

Statement. If $i\partial_t|\psi\rangle = H|\psi\rangle$ and $|\xi\rangle = U^\dagger|\psi\rangle$, then $i\partial_t|\xi\rangle = (U^\dagger H U - iU^\dagger\dot{U})|\xi\rangle$.

Smallest example. One spin with $H = \frac{\omega}{2}\sigma_z$, and a frame $U(t) = e^{-i\omega' t\sigma_z/2}$ that turns about z at the rate ω'. Here U commutes with H, so $U^\dagger H U = \frac{\omega}{2}\sigma_z$, and $-iU^\dagger\dot{U} = -\frac{\omega'}{2}\sigma_z$. The frame generator is

$$\mathcal{K} = \frac{\omega - \omega'}{2}\,\sigma_z.$$

For $\omega' = \omega$, $\mathcal{K} = 0$: a frame that turns with the precession sees the spin at rest. The term $-iU^\dagger\dot{U}$ is the price of a moving frame, a fictitious field along the rotation axis of the frame.

Used in 1.C.1, 2.C.1, 2.D.2.

Slater determinants and excitations (Problem 2, Useful information)

Statement. A Slater determinant is a state of independent fermions in chosen orbitals; a single (double) excitation replaces one (two) of its occupied orbitals.

Smallest example. Two fermions in four orbitals $1, \ldots, 4$, with reference $|1\,2\rangle$ (orbitals 1 and 2 occupied). Single excitations: $|3\,2\rangle$, $|4\,2\rangle$, $|1\,3\rangle$, $|1\,4\rangle$ (four). Double excitation: $|3\,4\rangle$ (one). Together with the reference, $1 + 4 + 1 = 6 = \binom{4}{2}$: every two-fermion state belongs to one of the three classes. A one-body operator $d_a^\dagger d_b$ moves one fermion, so it connects $|1\,2\rangle$ to single excitations only, never to $|3\,4\rangle$.

Used in 2.B.1, 2.C.1.

Operators on different spins (Problem 2, Useful information)

Statement. $\sigma_a\sigma_b = \delta_{ab}\mathbb{1} + i\epsilon_{abc}\sigma_c$; operators on different spins commute.

Smallest example. Two spins. In $[\sigma_1^x\sigma_2^x, \sigma_1^z]$ the factor σ_2^x commutes with σ_1^z and can be taken out:

$$[\sigma_1^x\sigma_2^x, \sigma_1^z] = [\sigma_1^x, \sigma_1^z]\,\sigma_2^x = -2i\,\sigma_1^y\sigma_2^x.$$

Only the spin that both operators act on produces a commutator; the result acts on the union of their spins.

Used in 2.B.2.

Exponentials of operators (Problem 2, Part C and the hint to C.1)

Statement. $e^{-S}\partial_t e^S = \int_0^1 e^{-uS}\dot{S}e^{uS}\,\mathrm{d}u$, and $e^{-S}\mathcal{K}e^S = \mathcal{K} + [\mathcal{K}, S] + \frac{1}{2}[[\mathcal{K}, S], S] + \cdots$.

Smallest examples. (i) If $S(t) = s(t)A$ with a fixed operator A, then S and $\dot{S}$ commute, the integrand equals $\dot{S}$ for every u, and the identity is the chain rule $e^{-S}\partial_t e^S = \dot{S}$. The integral matters only when S and $\dot{S}$ do not commute. (ii) One spin, $\mathcal{K} = \sigma_z$ and $S = -i\frac{\theta}{2}\sigma_x$. Then $[\mathcal{K}, S] = \theta\sigma_y$, $[[\mathcal{K}, S], S] = -\theta^2\sigma_z$, $[[[\mathcal{K}, S], S], S] = -\theta^3\sigma_y$, and the series sums to

$$e^{-S}\sigma_z e^S = \sigma_z\cos\theta + \sigma_y\sin\theta,$$

that is, σ_z turned towards σ_y by the angle θ.

Used in 2.C.1.

The trace norm and two inequalities (Problem 3, Useful information)

Statement. $|\operatorname{Tr}(AB)| \le \|A\|_1\,\|B\|$ and $\|\rho - |a\rangle\langle a|\|_1 \le 2\sqrt{1 - \langle a|\rho|a\rangle}$.

Smallest example. One qubit, $|a\rangle = |0\rangle$, and $\rho = |\psi\rangle\langle\psi|$ with $|\psi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$, $0 \le \theta \le \pi/2$. The matrix

$$\rho - |0\rangle\langle 0| = \begin{pmatrix} -\sin^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{pmatrix}$$

has eigenvalues $\pm\sin\theta$, so $\|\rho - |0\rangle\langle 0|\|_1 = 2\sin\theta = 2\sqrt{1-\cos^2\theta}$: for a pure ρ the second inequality is an equality. With $B = \sigma_z$ ($\|\sigma_z\| = 1$), $\mathrm{Tr}[(\rho - |0\rangle\langle 0|)\sigma_z] = -2\sin^2\theta$, and indeed $2\sin^2\theta \le 2\sin\theta$. For the mixed state $\rho = \mathbb{1}/2$, $\|\rho - |0\rangle\langle 0|\|_1 = 1 < 2\sqrt{1/2} \approx 1.41$: the inequality is strict.

Used in 3.A.2.

The partial trace (Problem 3, Useful information)

Statement. $\rho_X = \mathrm{Tr}_{\bar{X}}\,\rho$; if O_X acts only on X, then $\mathrm{Tr}(\rho O_X) = \mathrm{Tr}_X(\rho_X O_X)$.

Smallest example. Two qubits, X = qubit 1. In components, $(\rho_X)_{aa'} = \sum_b \rho_{ab,a'b}$. For the product state $|0\rangle \otimes |+\rangle$, $\rho_X = |0\rangle\langle 0|$ is pure. For $(|00\rangle + |11\rangle)/\sqrt{2}$, $\rho_X = \mathbb{1}/2$ is mixed, although the whole state is pure. With $O_X = \sigma_z \otimes \mathbb{1}$: $\mathrm{Tr}(\rho O_X) = \mathrm{Tr}_X(\rho_X \sigma_z)$ equals 1 for the product state and 0 for the entangled one.

Used in 3.A.1, 3.A.2, 3.A.3.

Commuting projectors (Problem 3, Useful information)

Statement. For commuting projectors, $1 - \prod_{x\in X}(1 - q_x) \le \sum_{x\in X} q_x$.

Smallest example. Two qubits, $q_x = |1\rangle\langle 1|$ on qubit x. The left side is $q_1 + q_2 - q_1 q_2$, the projector onto "at least one qubit is in $|1\rangle$". On $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ both sides are diagonal, with values $0, 1, 1, 1$ on the left and $0, 1, 1, 2$ on the right. They differ only on the doubly flipped state, which the sum counts twice.

Used in 3.A.2.

End of solution booklet